NMC 2011 A-LEVEL PAPER 1 QUESTIONS

SECTION A : 5 marks each

1. Let a and b be positive numbers not equal to 1. What is the value of $(\log_a b^2)/(\log_{a^2} b)$?
2. The sum of the first five terms of an arithmetic sequence is 40 and the sum of the first ten terms of the sequence is 155. What is the sum of the first 15 terms of the sequence?
3. How many real solutions does the equation $\left(\frac{2x^2-5}{3}\right)^{x^2-3x} = 1$ have?
4. Given that $\frac{a}{b+c+d} = \frac{4}{3}$ and $\frac{a}{b+c} = \frac{3}{5}$, find $\frac{d}{a}$?
5. Suppose that x and y are two real numbers such that $x - y = 2$ and $x^2 + y^2 = 8$, find $x^3 - y^3$.
6. Suppose John is given 1 penny on the first day, 2 pennies on the second day, 4 pennies on the third day, 8 pennies on the fourth day and so on for 18 days while David is given $\$1,000$ per day for 18 days. If $\$1 = 10$ pennies, who will have the larger amount of money at the end of 18 days. Give your answer in dollars ($).
7. When Mum came home and found the cookie jar broken, she called all her four children for an explanation and the following discussion took place:
   Ann: I did not break the jar.
   Bob: I did not break the jar and Ann did not break the jar.
   Cal: I did not break the jar and Bob did not break the jar.
   Deb: Ann did not break the jar and Bob did not break the jar.
   Mum later found out that only one of the children lied. Which one of the children broke the cookie jar?
8. Let K be the set consisting of the first 11 positive integers. A subset of three distinct positive integers is said to be *acceptable* if it contains at most one odd element. If a subset of three different numbers is chosen at random from K, what is the probability that the subset will be *acceptable*?
9. Suppose that $a > 0$. If $x^{a/2} - 4 + 4x^{-a/2} = 0$, find the reciprocal of $\frac{1}{2} + x^a$.
10. Two ladders, one of which is twice as long as the other, rest on the floor and reach the same vertical height on the wall. The shorter ladder makes an angle of 60° with the floor. What angle does the longer ladder make with the floor?

SECTION B : 10 marks each

11. Suppose that f is a real-valued function whose domain is the set of positive integers and that f satisfies the following two properties

$$\begin{aligned} f(1) &= 23 \text{ and} \\ f(n+1) &= 8 + 3f(n) \text{ for all } n \geq 1. \end{aligned}$$

It follows that there are constants a, b and c such that $f(n) = a \cdot b^n - c$ for $n = 1, 2, \ldots$ Find the value of $a + b + c$.

12. (a) How many real numbers p are there so that the polynomial $x^{10} + px + 1 = 0$ has a real solution S and also has $\frac{1}{S}$ as a solution?

(b) Given that $T = 1997^2 - 1996^2 + 1955^2 - 1994^2 + \ldots + 3^2 - 2^2 + 1^2$, find the largest prime factor of T.

13. Given that $\cos 5x = A\cos^5 x + B\cos^4 x + C\cos^3 x + D\cos^2 x + \cos x + F$, find the sum of A, B, C, D, E and F.

14. In a magic square, each row, column, and diagonal has the same sum. All the odd numbered years from 1987 to 2003 inclusive are used to form the magic square without repeating any year. Given the locations of 1999 and 2003, what is the year that belongs in the box marked X?

	1999	
2003		
		X

15. In the diagram below, point P is chosen on diameter $\overline{AC}$ of a circle so that $\overline{PC} = 3$ *cm*. The area of the circle is 64π cm^2 and chord AB is 12 *cm* long. Compute the distance from P to B.

